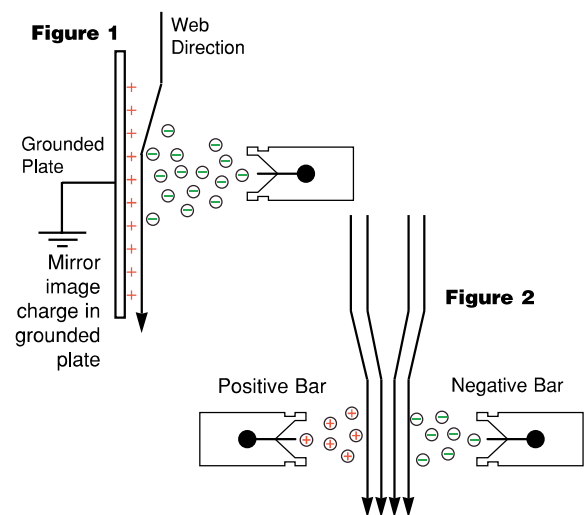


STATIC GENERATION EQUIPMENT



What is Static Generation?

Static Generation creates a controlled static charge on a non-conductive material that will allow a temporary, adhesion between surfaces with charges of opposite polarity. A high DC voltage (up to 50kV) is safely produced and is carried to a static generator bar, where it is connected to an array of suitably designed emitter pins to generate a “corona”. The DC voltage may be positive or negative. The emitter pins are positioned within close proximity of a grounded surface (fig.1) or, for greater effect, a generator bar with emitter pins of the opposite polarity (fig.2). The material(s) to be bonded are passed into the “corona”, resulting in bonding to the grounded surface or to the other material.



Static Generation Applications

Popular Application Example	Product														
		Drinks	Food	Product Packaging	Conveying & Product Handling	Thermofforming	Plastic Rigid	Blow Moulding	Plastics Film Extrusion	Printing Sheet	Printing Web	Printing Other	Print Finishing	Converting Paper	Textiles
		Food and Drinks	Rigid Plastics	Flexible Plastics	Printing	Converting	Pharma / Medical	Other							
CD & DVD Delamination During Manufacture	993R Spark Free Generator Bar														
Static Pinning On Metal Or Glass	993R Spark Free Generator Bar														
Static Pinning To Web Based Materials	993R Spark Free Generator Bar														
Turret Re-Winders	993R Spark Free Generator Bar, 992v3 30kV Static Generator														
Manufacture Of Carrier Bags	993R Spark Free Generator Bar, 992v3 30kV Static Generator														
Ribbon Tacking In Newspaper And Magazine Web Converting	993R Spark Free Generator Bar, 992v3 30kV Static Generator														
Static Pinning Gusset Into Flat Bag	993R Spark Free Generator Bar, 992v3 30kV Static Generator														
Static Pinning Of Web To Core On Turret Rewind	993R Spark Free Generator Bar, 992v3 30kV Static Generator														
Static Pinning On Side Weld Bag Handles	993R Spark Free Generator Bar, 992v3 30kV Static Generator														
IML Decoration Of Plastic Moulds	994IML Hydra, 993R Spark Free Generator Bar														
Static Pinning Film On To A Cast Film Extrusion Chill Roller - 995 Claw	995 Claw Style Edge Pinning, 992v3 30kV Static Generator														
Reduced Material Overlap On Horizontal Over Wrapper	995R Handheld IML Pinning Head														
Stacking & Pinning Food Sachets Before Wrapping	995R Handheld IML Pinning Head														
Board Laminating	Generator Bars														
Chocolate Sticking To Moulds During Chocolate Manufacture	Generator Bars														
Controlling Fire Risk During Processing Of Sugar	Generator Bars														
Controlling Static Charges During Chocolate Production	Generator Bars														
Operator Shocks During Chocolate Packaging	Generator Bars														

Static Generators

Meech Generators and generator bars are designed to create a controlled level of static charge to impart temporary bonding between materials.



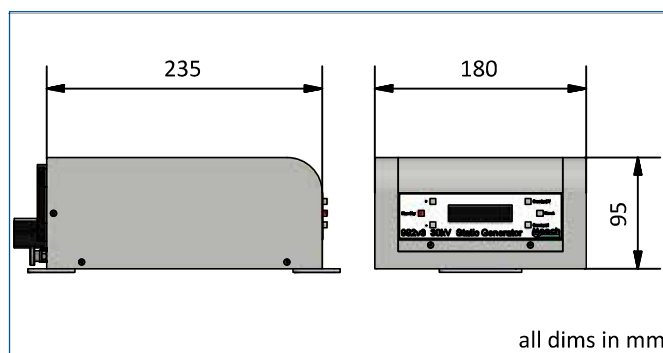
	992v3-30 Generator	992v3-50 Generator	994 IML Static Generator	Hyperion 994CG Generator
Weight	3.25kg	3.25kg	500g	400g/350g
Dimensions (HxWxD)	235 x 180 x 95mm	235 x 180 x 95mm	190 x 171 x 45 mm	45 x 40 x 150mm (Socket) 45 x 40 x 170mm (Terminal)
Construction	Painted Steel Enclosure	Painted steel enclosure	DuraForm PA (Nylon 12)	DuraForm PA (Nylon 12)
Maximum Temperature	60°C	60°C	45°C	55°C
Power Input	Mains supply of 85 to 265V, 47 - 63Hz	Mains supply of 85 to 265V, 47 - 63Hz	24VDC	21-30V DC*
Output Polarity	Dedicated positive or negative	Dedicated positive or negative	Negative	Negative
Output Voltage	0-30kV	0 to 50kV	0 - 20kV Negative	4-25kV
Maximum Output Current	0 to 500µA	0 to 260µA	500µA	500µA
Input Current				200mA-850mA Load Dependant
Settings Control	Front Panel	Front Panel	Analogue / Digital Input	BarMaster, PLC Control or Remote Set Point Control
Environmental Protection				IP66 – Socket IP64 – Terminal
Connection	IEC socket	IEC socket	8 Way mini DIN 2 x HT output sockets	5 Pin M12
Scan QR Code for Operating Manual				



992V3 30kV & 992v3 50kV Static Generator

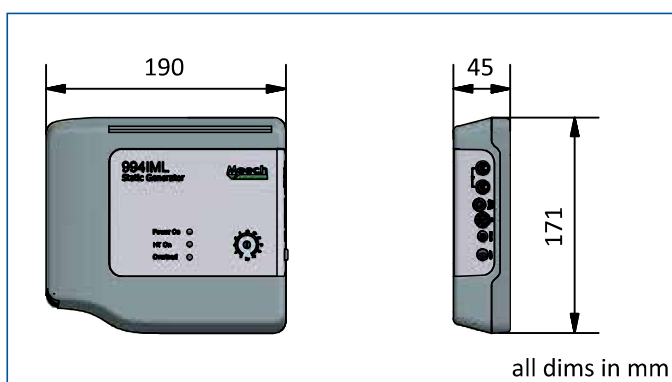
The 992v3 provides a high voltage output adjustable over the range of either 0-30kV or 0-50kV or a constant current can be adjusted between 0 and 500 μ A or 0 and 250 μ A. The 992v3 is used in conjunction with the 993R spark free generator bar and is available in either negative or positive output polarity versions.

- Available in two versions.
- Selectable constant voltage or constant current mode.
- Auto trip on detection of flashover, which reduces the risk of damage to equipment and/or materials.
- Internal switch mode power supply that can accommodate different input voltages and frequencies.
- Compact size.
- Twin outputs meaning multiple Generator bars can be connected.



994 IML Static Generator

The 994-IML Static Generator is a compact and lightweight system, it provides 0-20kV, Negative polarity with connections to suit the Meech Hydra.

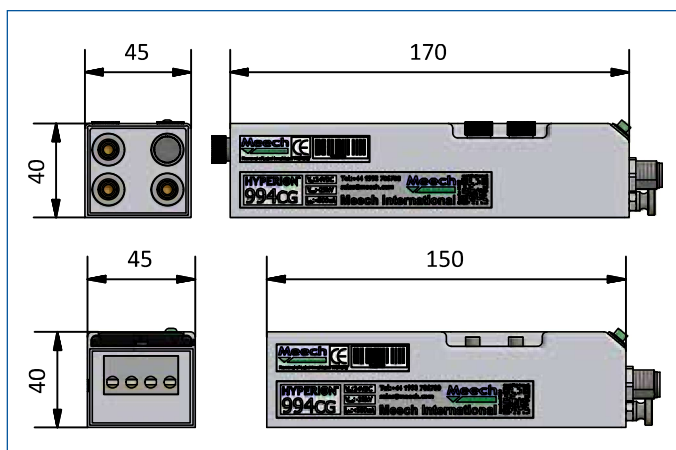


- Adjustable 0-20kV output to suit all IML applications.
- Lightweight allowing it to be mounted on a robot.
- 24V DC Supply.
- Current limiting circuitry that prevent damage to the unit.
- Full remote control for easy integration with PLC.

Hyperion 994CG Compact Generator

HYPERION™
A Meech Innovation

The 994CG is the most powerful and compact IML generator Meech have ever produced. Powered by 24V DC, it has outlet ports capable of connecting up to 4 x 994 Hydra IML Pinning Systems directly to the generator. The 994CG is available in two versions, a socket terminal utilises the Meech PVC plug and a terminal version that allows crimp connections.



- Compact design.
- Adjustable output voltage that can be programmed remotely with a PLC control or a Meech BarMaster
- PLC Control; the high voltage from the 994CG can be remotely adjusted via the robot PLC.
- External Set Point Control; a voltage adjuster can be installed into the system via a break-in connector.
- Connect up to 4x 994Hydra IML pinning systems



Static Generation Bars & Heads

The Meech In-Mould Labelling systems use static electricity to charge a label while it is being transferred to the moulding machine. All Meech systems are lightweight and easily installed.



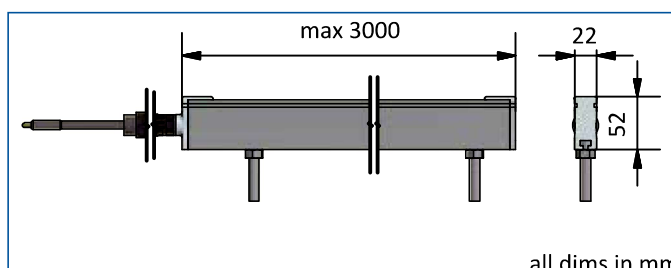
	993R	995R	995v3	995CLAW	995R IML	995DE
Dimensions (WxHxL)	22mm x 52mm max length 3000mm	4P2R 65x30x50mm 8P2R 107x30x50mm 12P2R 144x25x50mm	96 x 26 x 300mm	30 x 65 x 154mm	30 x 65 x 155mm	56 x 56 x 142mm
Cable	2 metre	2 metre	2 metre	5 metre	3 metre	5 metre
Max temperature	60°C	115°C				
Operating voltage	Up to 50kV DC					
Construction	PTFE extrusion					
Emitter Pins	Titanium					
Power Source	Meech 992v3 Static Generator					
Operating current	500uA					
Operating polarity	Either +VE or -VE					



993R Spark Free Generator Bar

The 993R is a high performance generator bar used with the Meech range 992V3 30kV and 50kV Static Generators. Resistively coupled emitter pins deliver smooth, controlled pinning whilst spark free operation avoids tripping out on intermittent applications.

- Greatly improved operator safety.
- Removable emitter pin option Optimum performance can be maintained. Life of the bar is substantially lengthened.

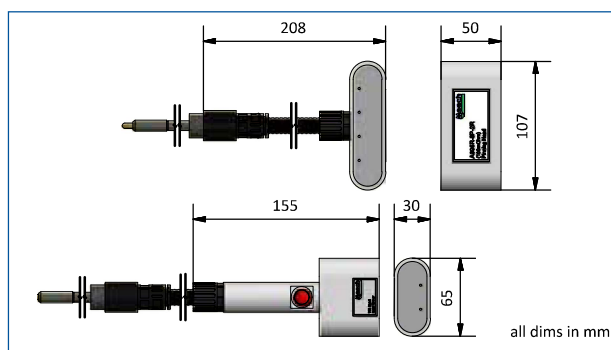


995R & 995R IML Pinning Head

The 995 is designed to deliver exceptional pinning of multi-layer plastic. Uses include bag making and flow-wrapping machinery. The integral switch on the 995R IML allows the operator to switch the system on only when pinning is required. Ideal for manual In Mould Labelling applications.

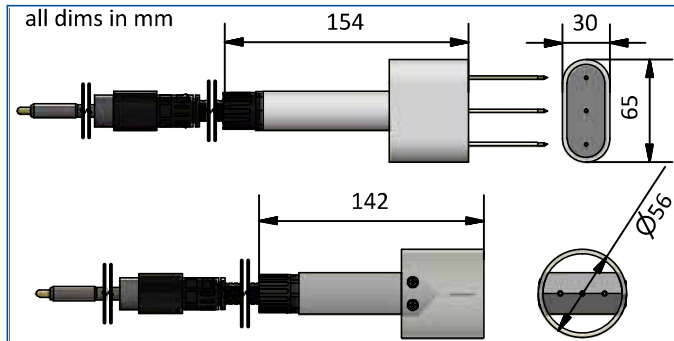


- Resistively coupled allowing spark free, independent operation.
- High pin density resulting in powerful pinning.



995 DE & 995 Claw Pinner

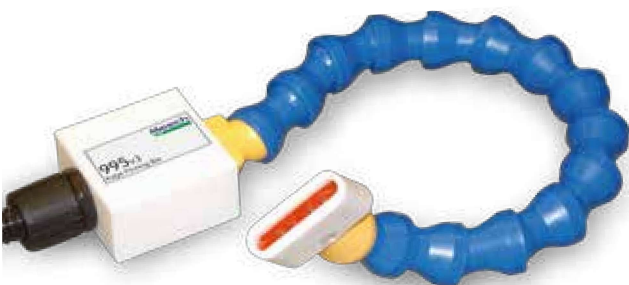
The 995 DE Dirty Environment Pinner provides electrostatic pinning in applications where the risk of contamination is high. The resistively-coupled titanium pins are shrouded to offer a physical protection. The 995 Edge Pinning Claw is a powerful pinning head intended for use on cast-film lines. Pinning the edges of the cast film to the chill roller prevents necking of the film.



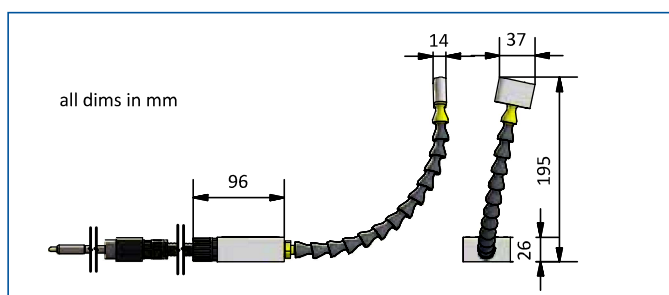
- Resistively coupled allowing spark free, independent operation.
- High pin density resulting in powerful pinning.
- Open design prevents build-up of contamination.
- Flexible titanium pins that can be adjusted to suit the specific application.

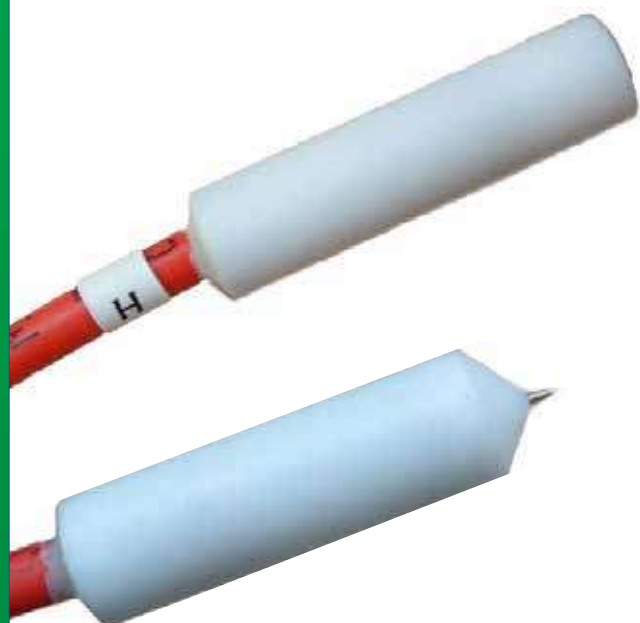
995v3 Flexible Pinning Head

The 995v3 Flexible Pinning Head Includes a length of flexible trunking that allows accurate pinning of non-conductive materials. Ideal for hard to reach applications.



- Resistively coupled allowing spark free, independent operation.
- Stayput flexible trunking allows the Pinning head to be easily positioned.



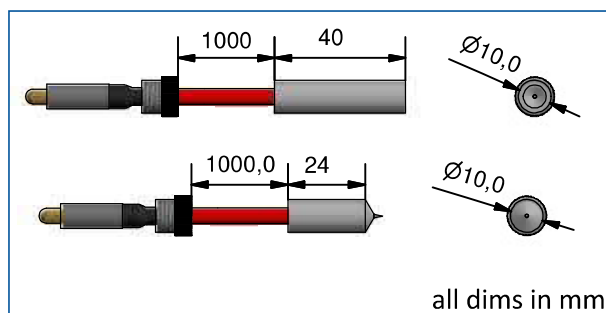


- Resistively coupled allowing spark free independent operation.
- High pin density resulting in powerful pinning.

994 SPP Heads

The 994 Single Point Pinning Heads are ideal for localised static pinning in applications such as in-mould labelling (IML).

The pinning heads are connected to a 992v3 high voltage supply via a resistive splitter box (4 way or 6 way). The individual resistive coupling ensures smooth spark-free pinning, with completely independent operation of each head.



Options for Static Generation Equipment



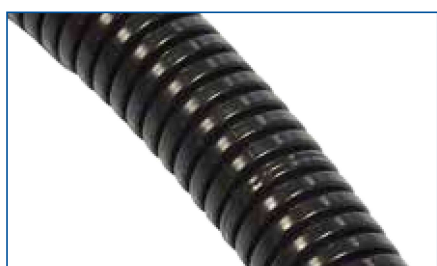
994 4 or 6 Way Standard Resistive Splitter Box

The splitter box can be used to simultaneously run multiple 994 SPP Heads or 994 Hydras from a single power supply. This is a safe way to increase the amount of pinners from a single source. The splitter box is available with 4 or 6 splitting outlets. The splitter box has a resistance of 500MΩ.



994 4 or 6 Way Heavy Duty Resistive Splitter Box

The heavy duty resistive splitter box has a resistance of 100MΩ, which offers a more powerful output. The heavy duty splitter box is available with 4 or 6 splitting outlets.



HT Cable with Conduit for 993R bars

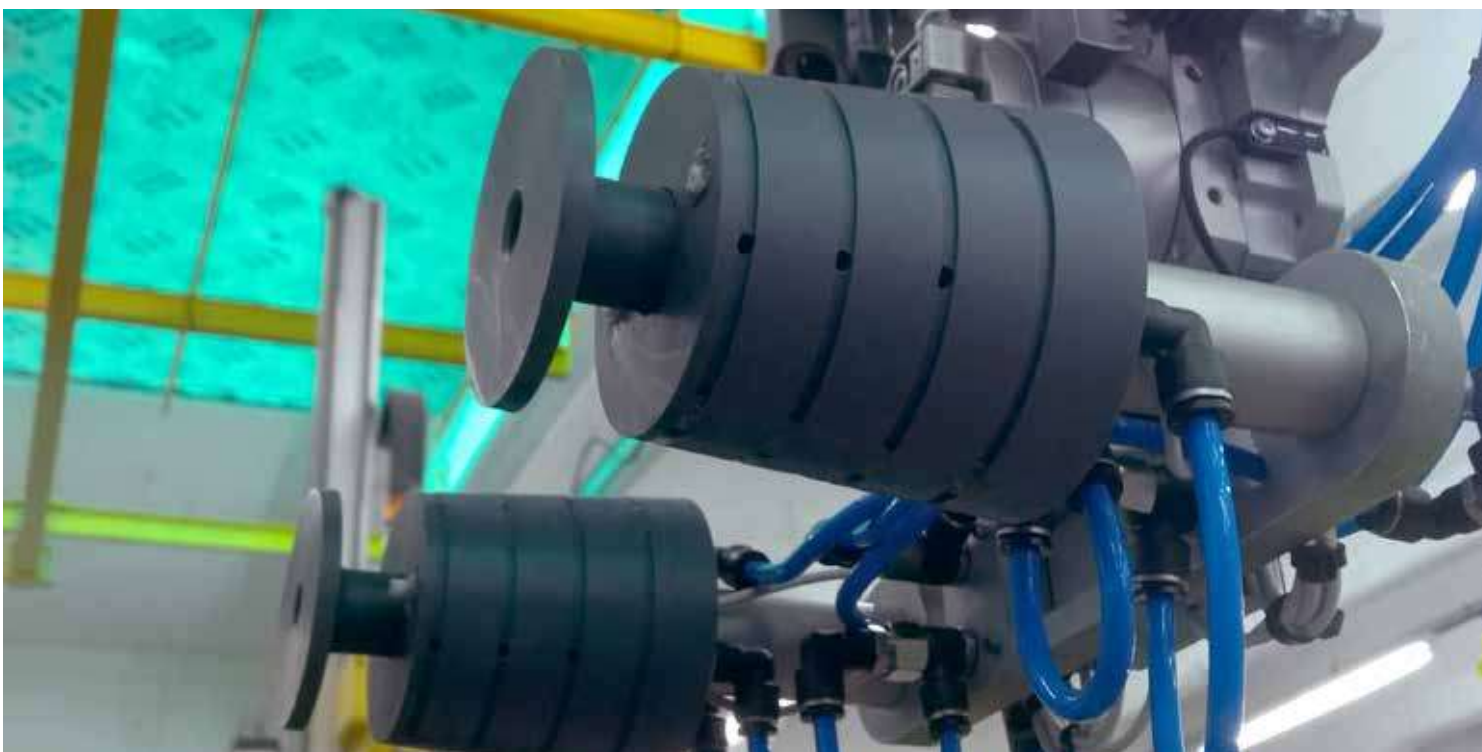
As standard, the 993R bar is provided with 2m of HT Cable. On purchase of the bar, HT cable can be increased to lengths of up to 20m.

IML Hydra Pinning Systems

In Mould Labelling requires the use of pinning systems to create the perfect finish. Meech pinning systems offer the ultimate in controlled static generation, ideal for the IML industry.



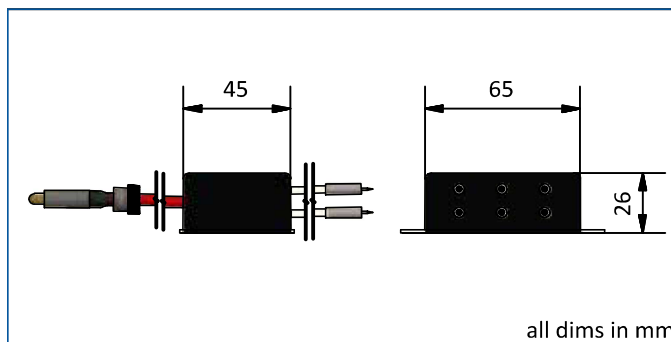
	994 IML Hydra Socket Flanged	994 IML Hydra Socket Flangeless	994 IML Hydra Terminal Flanged	994 IML Hydra Terminal Flangeless	994 IML Hydra Distributor
Dimensions (LxWxH)	42 x 30 x 19mm	27 x 40 x 15mm	42 x 30 x 19mm	27 x 40 x 15mm	45 x 65 x 26mm
Input Cable	500mm	500mm	500mm	500mm	1000mm
Pinner Cable Length	300mm	300mm	300mm	300mm	450mm
Weight	90g-130g				175g
Operating voltage	Up to 25kV DC				Up to 20kV
Construction	Pinning Head - PTFE Distributor - ABS				
Emitter Pins	Titanium				
Max temperature	Pinning Head - 105°C Distributor - 90°C				
Power Source	994CG Socket or 994IML		994CG Terminal		994CG Socket or 994IML
Pinning Heads Number	2/4/6	2/4/6	2/4/6	2/4/6	4/6





994 Hydra IML Pinning System

994 Hydra IML Pinning System is a pinning system designed for applications where small items are being moulded i.e yoghurt pots or drinks containers. The standard system comprises a resistively coupled distributor with either four or six pinner cables.

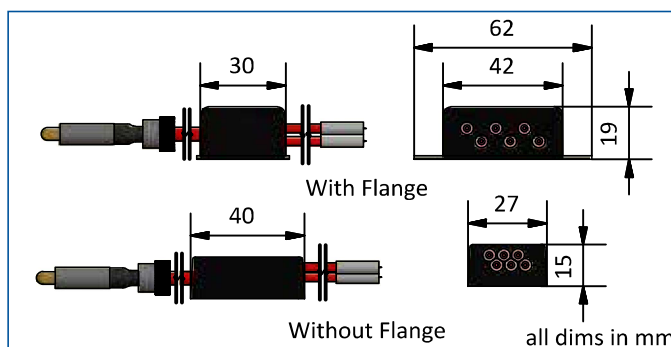


- Miniature pinning head (5 x 15mm) that fits small moulds.
- Resistive coupling offers smooth, spark-free pinning, with independent operation of each pin.
- Small distributor housing allows it to be easily mounted on tool.



994 Hydra Miniature IML Pinning System

The 994 Hydra is a miniaturised pinning system designed for applications where small items are being moulded i.e yoghurt pots or drinks containers. The standard system comprises a resistively coupled distributor with either two, four or six pinner cables.



- Miniature pinning head (5 x 15mm) that fits small moulds.
- Resistive coupling offers smooth, spark-free pinning, with independent operation of each pin.
- Small distributor housing allows it to be easily mounted on tool.

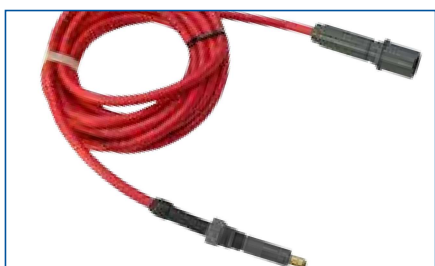


Options for IML Hydra Pinning Systems



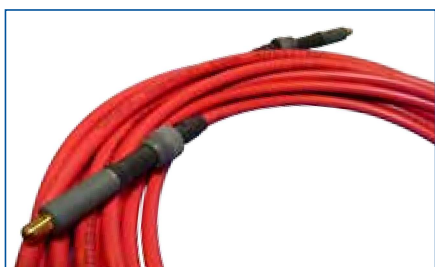
994 Hydra IML Splitter

Available in 2, 4, 6 and 8 splitter configurations. Can be used to power additional 994 Hydras from a single source for IML applications.



Hydra HT Supply Cable 5000mm Male - Female

Additional extension cable to be used for direct plug-in from the Generator to the Hydra.



Hydra HT Supply Cable 5000mm Male - Male

Additional extension cable to be used for direct plug-in from the Generator to the Hydra Splitter.



994CG Remote Setpoint Controller

Generates an adjustable 4-20mA signal to control the output voltage of the 994CG Compact Generator. M8 input and M12 output connections.



M12-M8 Murr adaptor cable 1000mm for BarMaster to 994CG connection

Allows connection of BarMaster programmer to 994CG Compact Generator



Mini Din prewired cable 8 way

Compatible with the I/O ports on 977cm and 994IML. The pre-wired cable simplifies the connection to input and output signals.